

UST-MAYA, Russian Federation

WMO: 249660

Lat: 60.3650N

Lon: 134.4510E

Elev: 170

StdP: 99.30

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.0	-48.4	-51.4	0.0	-49.8	-50.0	0.0	-48.4	4.2	-27.0	3.3	-29.6	0.1	20	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	29.4	19.1	27.5	18.2	25.5	17.2	20.3	26.8	19.2	25.4	18.1	23.9	2.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	22.9	16.9	12.3	22.2	15.8	11.4	21.0	59.5	26.9	55.7	25.4	52.1	23.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.0	4.2	-51.6	32.5	1.8	1.5	-52.9	33.5	-54.0	34.4	-54.9	35.3	-56.2	36.4		
(5)			-51.7	22.0	1.8	1.3	-52.9	22.9	-54.0	23.6	-54.9	24.4	-56.2	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.3	-38.7	-33.4	-19.0	-3.3	7.9	15.9	18.7	15.2	6.5	-6.5	-26.2	-38.3		
	DBStd	21.75	6.21	6.41	7.35	5.83	3.96	3.80	3.52	3.71	4.22	6.55	9.09	6.64		
	HDD10.0	7334	1510	1214	898	399	89	3	0	4	118	513	1087	1498		
	HDD18.3	9805	1768	1447	1157	649	324	91	39	111	353	771	1337	1757		
	CDD10.0	652	0	0	0	0	23	179	269	165	15	0	0	0		
	CDD18.3	81	0	0	0	0	0	17	50	14	0	0	0	0		
	CDH23.3	937	0	0	0	0	8	253	539	136	1	0	0	0		
	CDH26.7	241	0	0	0	0	1	56	160	24	0	0	0	0		
	WSAvg	1.6	0.6	0.8	1.4	2.0	2.4	2.2	2.0	1.8	1.9	1.7	1.1	0.6		
(15) Precipitation	PrecAvg	309	11	9	8	10	28	36	55	50	42	28	20	11		
	PrecMax	429	25	23	43	32	55	71	137	142	88	65	40	24		
	PrecMin	187	3	2	1	1	4	10	6	9	11	10	6	3		
	PrecStd	62	5	5	9	8	13	16	33	31	22	14	9	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-17.4	-10.6	3.3	13.3	24.4	30.7	32.3	29.3	21.2	8.0	-0.2	-18.8		
		MCWB	-17.8	-11.4	-0.3	6.3	13.6	19.0	20.6	19.7	13.6	4.4	-1.7	-19.2		
	2%	DB	-23.1	-16.3	-0.1	9.9	20.9	28.4	30.2	26.7	18.1	5.0	-4.1	-22.0		
		MCWB	-23.4	-16.8	-2.9	4.1	11.7	17.5	19.7	18.6	12.2	2.8	-4.7	-22.2		
	5%	DB	-26.8	-19.8	-3.2	7.7	18.4	26.4	28.5	24.6	15.6	2.9	-8.6	-25.3		
		MCWB	-27.0	-20.2	-5.4	2.6	10.3	16.5	19.1	17.2	10.6	1.0	-9.3	-25.5		
	10%	DB	-29.7	-23.6	-6.1	5.5	16.1	24.3	26.5	22.5	13.7	1.2	-13.0	-28.6		
		MCWB	-29.8	-23.8	-7.8	1.2	8.9	15.7	18.2	16.1	9.7	-0.3	-13.4	-28.8		
	0.4%	WB	-17.7	-11.1	0.1	6.4	13.9	20.0	22.1	21.1	14.6	5.2	-1.8	-19.2		
		MCDB	-17.4	-10.5	3.3	12.7	23.5	28.0	29.6	26.6	19.3	7.0	-0.7	-19.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-23.3	-16.7	-2.9	4.2	12.1	18.6	21.0	19.4	13.1	2.8	-4.7	-22.2		
		MCDB	-23.2	-16.3	-0.2	9.2	20.2	26.1	27.8	24.7	16.8	4.8	-4.1	-21.9		
	5%	WB	-27.0	-20.1	-5.4	2.9	10.7	17.5	20.0	18.1	11.7	1.2	-9.3	-25.5		
		MCDB	-26.9	-19.8	-3.3	7.2	17.7	24.3	26.3	23.2	14.6	2.6	-8.7	-25.3		
	10%	WB	-29.7	-23.8	-7.9	1.5	9.3	16.4	18.9	16.8	10.2	-0.1	-13.4	-28.7		
		MCDB	-29.6	-23.5	-6.2	5.2	15.2	23.0	25.0	21.6	13.1	1.1	-13.0	-28.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	11.7	16.5	12.7	11.1	12.2	11.5	10.9	9.6	7.0	7.7	5.7		
		MCDBR	9.4	12.9	16.6	12.7	15.9	16.0	15.1	14.0	13.4	7.9	9.7	8.0		
		MCWBR	9.2	12.4	14.0	7.9	8.5	6.9	6.3	6.8	7.9	5.6	9.2	8.0		
	5% WB	MCDBR	9.5	12.9	15.9	11.7	14.5	14.1	12.4	12.4	11.0	7.3	9.7	8.0		
		MCWBR	9.3	12.4	13.4	7.5	7.9	6.6	5.7	6.4	7.0	5.4	9.3	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.224	0.263	0.309	0.384	0.453	0.430	0.517	0.458	0.350	0.300	0.236	0.196		
	taud		2.007	2.076	2.040	1.961	2.030	2.222	1.959	2.096	2.391	2.276	2.019	1.925		
	Ebn at Noon		528	723	799	794	767	800	706	721	768	678	493	397		
	Edh at Noon		45	77	113	149	151	127	162	131	82	64	43	30		
(44) All-Sky Solar Radiation	RadAvg		0.50	1.56	3.31	5.08	5.18	5.94	5.42	4.21	2.57	1.21	0.64	0.27		
	RadStd		0.03	0.06	0.12	0.20	0.42	0.42	0.33	0.36	0.25	0.13	0.06	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (4 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-164	N/A		

Nomenclature: See separate page